



Setting Up Vite + React Projects

A comprehensive guide to project structure, npm commands, and development workflow for front-end developers

Understanding Your Project

Structure



src/ Directory

Core working directory containing all React code, components, and assets



Root Files

Configuration files including package.json, vite.config.js, and index.html



Dependencies

Node modules and package management through npm

The project structure follows modern best practices for React development with Vite, ensuring clean organization and maintainability.

Source Folder

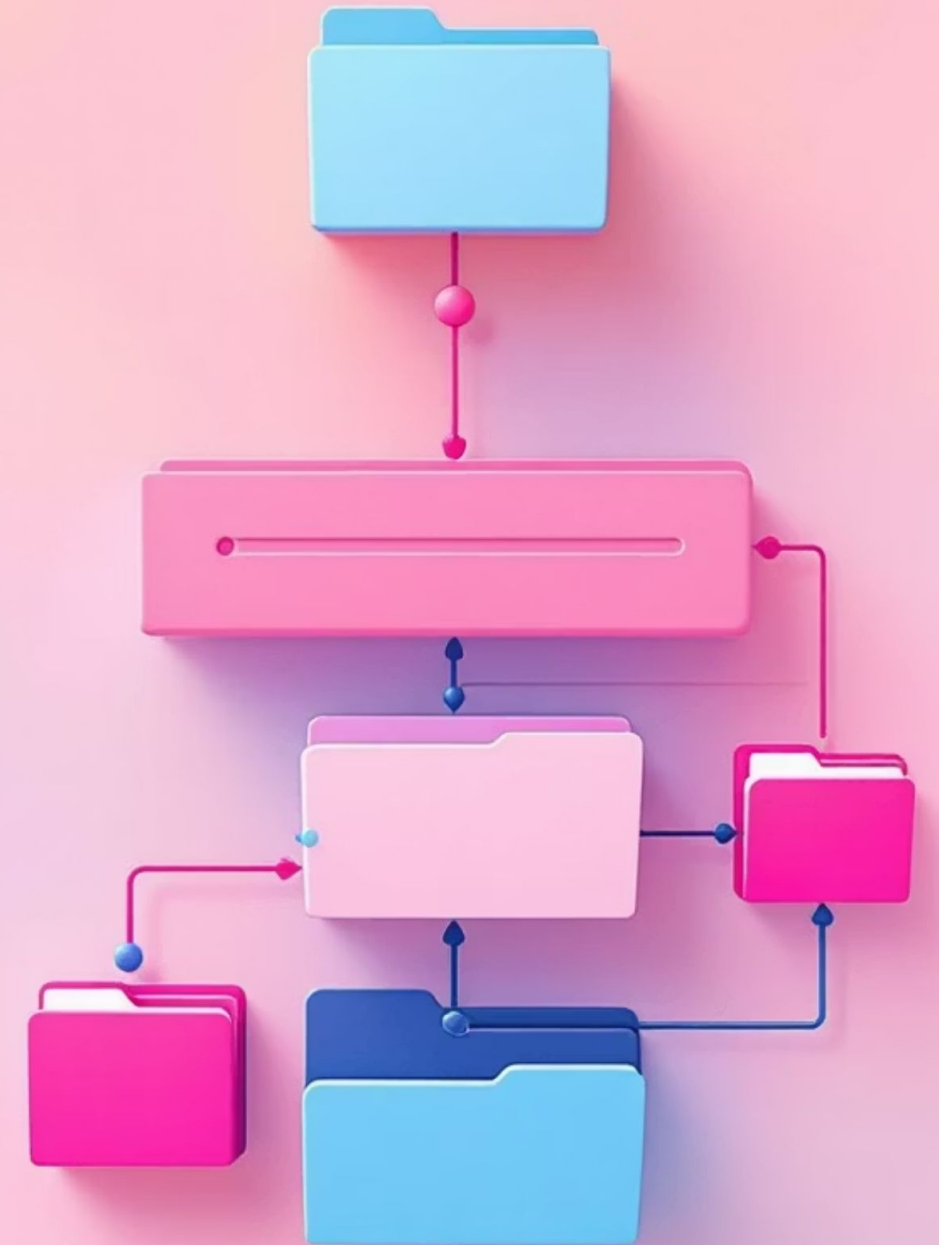
Structure Main Components

- main.jsx - Application entry point
- App.jsx - Root component managing UI
- App.css - Component-specific styles
- index.css - Global styles applied everywhere

This structure keeps your code modular and scalable as your application grows.

Organizational Folders

- assets/ - Images, icons, and fonts
- components/ - Reusable UI components
- utils/ - Helper functions and utilities



Root Directory Files

1

index.html

Single HTML file implementing SPA concept with root div for React rendering

2

package.json

Project configuration including dependencies, scripts, and metadata

3

package-lock.json

Automatically generated file locking dependency versions

4

vite.config.js

Vite-specific configuration for build and development settings

5

.gitignore

Specifies files to exclude from version control

6

eslint.config.js

Code linting and quality check configuration

Creating Your Project

The Command

```
npm create vite@latest
```

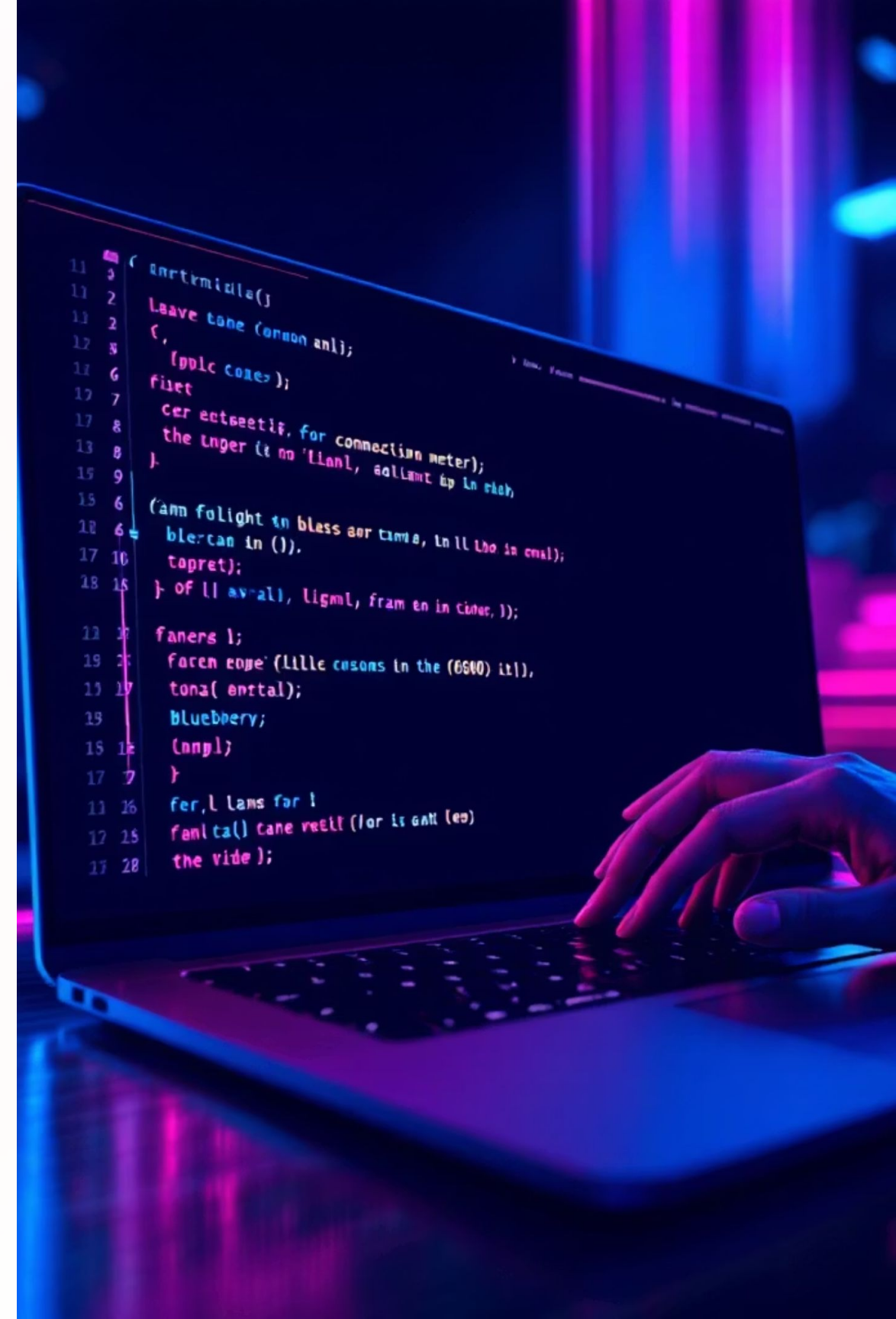
Breaking It Down

npm - Node Package Manager for JavaScript packages

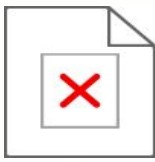
create - npm command to initialize projects

vite@latest - Latest version of Vite tool

This command scaffolds a new Vite project with React template, setting up all necessary configuration files and folder structure automatically.

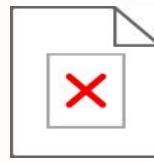


What is npm?



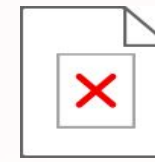
Package Manager

Installs, manages, and runs JavaScript packages. Comes bundled with Node.js installation.



Dependency Management

Handles project dependencies listed in package.json with version control.



Script Runner

Executes development commands like dev server, build processes, and testing.



What is Vite?

⚡ Fast Startup

Near-instant server startup
using native ES modules
instead of bundling everything

↺ Hot Module Replacement

Instant updates without page
reload when you modify code

⚡ Faster Than CRA

(Old version of React js => npm

~~create-react-app~~)

Outperforms Create React App with modern tooling and optimized
build process



Installing Dependencies

npm install



Reads package.json and installs all dependencies



node_modules

Creates folder containing all installed packages



Ready to Run

Project environment is fully prepared

Run this command after creating a new project or cloning an existing one to set up your development environment.

Starting the Development Server

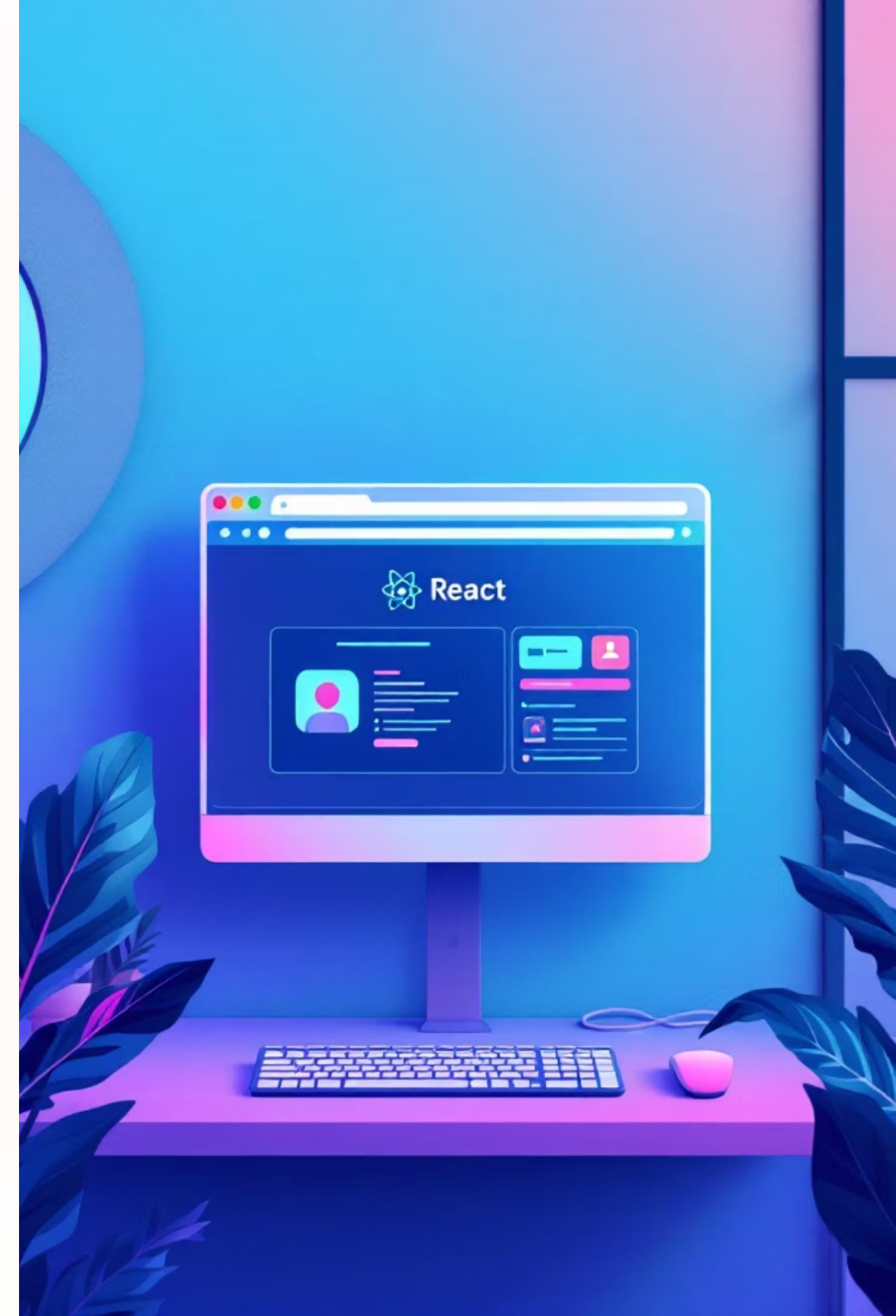
Running the Command

```
npm run dev
```

This starts the Vite development server and opens your project at <http://localhost:5173>

What It Does

- Launches local development server
- Enables live updates without reloading
- Serves your React application



Customizing Scripts

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¹ Edit package.json

Locate the "scripts" section in your configuration file

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³ Update Commands

Ensure the command still points to "vite" executable

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² Modify Script

Names
Change "dev" to your preferred name like "start" or "serve"

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⁴ Run New Script

Use npm run [your-script-name] to start the server



Key Rule: Script names are user-defined. The command format is always `npm run [script-name]`